

NHANES 2021-2023 Inferential Statistics Analysis - Answers

Data Sources Used

- **DEMO_L.xpt**: Demographics data (n=11,933)
- **BPXO_L_bloodpressure.xpt**: Blood pressure measurements (n=7,801)
- **PAQ_L_physicalactivity.xpt**: Physical activity questionnaire (n=8,153)
- **BMX_L_bodymeasures.xpt**: Body measurements (n=8,860)

Data Preparation Notes

- **Marital Status (DMDMARTZ)**: Recoded to binary - Married (1) vs. Not Married (0, including widowed, divorced, separated, never married, and living with partner)
 - **Education Level (DMDEDUC2)**: Recoded to binary - Bachelor's degree or higher (5) vs. Less than bachelor's (1-4)
 - **Sedentary Behavior (PAD680)**: Cleaned by removing placeholder values 7777 and 9999
 - **Weight**: Used measured weight (BMXWT) since self-reported weight (WHD020) was not in the provided dataset
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Question 1: Association between Marital Status and Education Level

Research Question: Is there an association between marital status (married or not married) and education level (bachelor's degree or higher vs. less than a bachelor's degree)?

Statistical Test Used

Chi-Square Test of Independence - This test is appropriate because both variables are categorical with two levels each, and we are testing whether there is a relationship between them.

Results

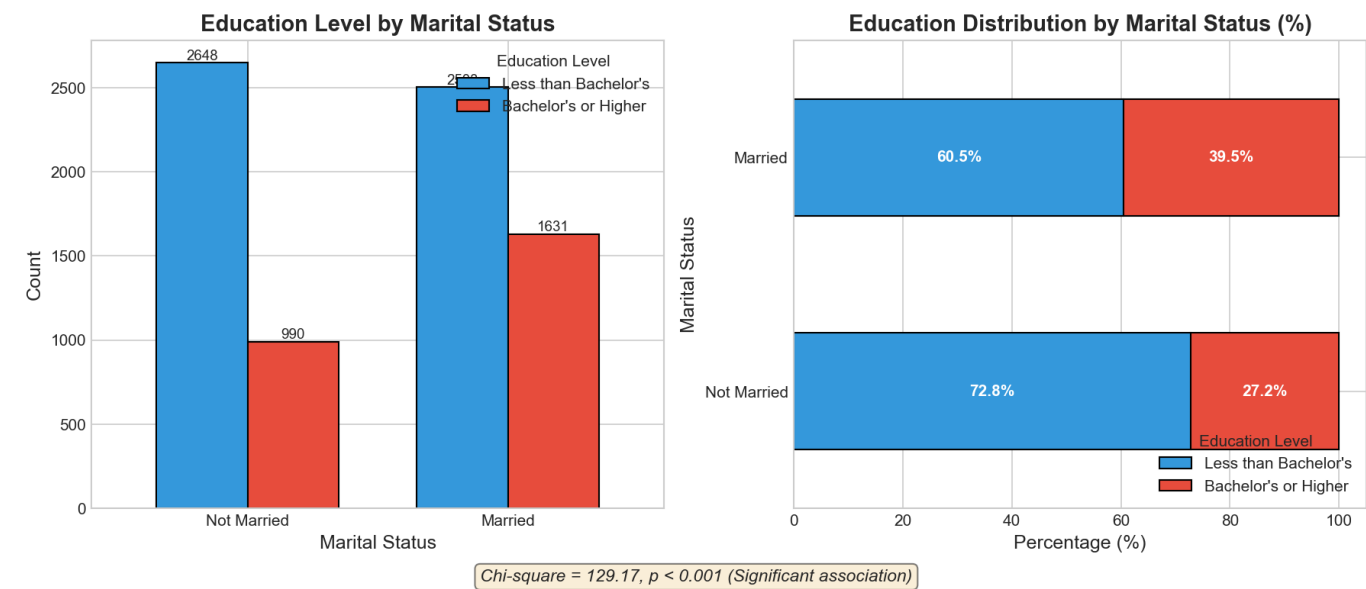
Contingency Table:

	Less than Bachelor's	Bachelor's or Higher
Not Married	2,648	990
Married	2,503	1,631

Statistical Output:

- Chi-square statistic: 129.17
- Degrees of freedom: 1
- **P-value: < 0.001**

Visualization



Interpretation

We REJECT the null hypothesis ($p < 0.001$).

There is a **statistically significant association** between marital status and education level. Looking at the proportions:

- Among **Not Married** individuals: 72.8% have less than a bachelor's degree, 27.2% have a bachelor's or higher
- Among **Married** individuals: 60.6% have less than a bachelor's degree, 39.5% have a bachelor's or higher

Conclusion: Married individuals are more likely to have a bachelor's degree or higher compared to non-married individuals. This could reflect various socioeconomic factors where education and marriage decisions may be interconnected (e.g., career stability, delayed marriage for education, or shared socioeconomic backgrounds in partner selection).

Question 2: Difference in Sedentary Behavior by Marital Status

Research Question: Is there a difference in the mean sedentary behavior time between those who are married and those who are not married?

Statistical Test Used

Welch's t-test (Independent Samples t-test with unequal variances) - This test is appropriate because we are comparing means of a continuous variable (sedentary minutes) between two independent groups (married vs. not married). Welch's version was used because Levene's test indicated unequal variances ($p < 0.001$).

Results

Descriptive Statistics:

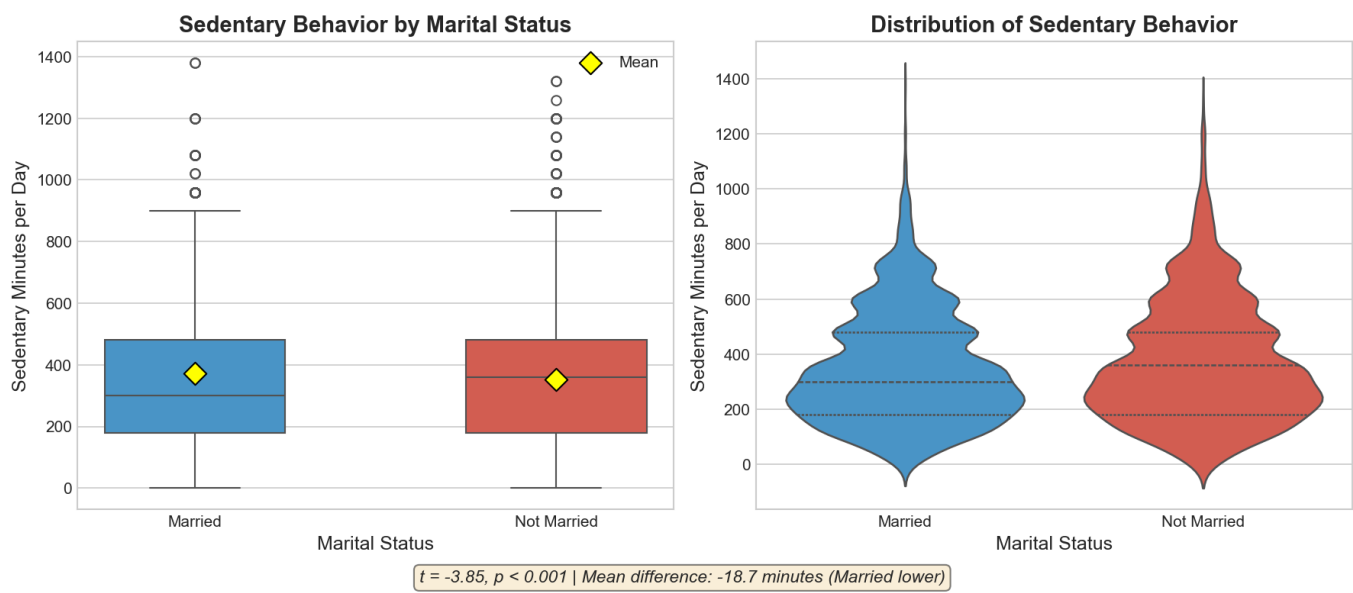
Group	N	Mean (min)	Std Dev	Median (min)
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Group	N	Mean (min)	Std Dev	Median (min)
Married	4,106	353.29	203.88	300
Not Married	3,603	371.96	219.53	360

Statistical Output:

- t-statistic: -3.85
- **P-value: 0.000118**
- Mean difference: -18.67 minutes (Married - Not Married)
- Effect size (Cohen's d): -0.088 (small effect)

Visualization



Interpretation

We **REJECT** the null hypothesis ($p < 0.001$).

There is a **statistically significant difference** in sedentary behavior between married and non-married individuals. On average, married individuals spend about **18.67 fewer minutes** per day in sedentary activities compared to non-married individuals.

However, the effect size is small (Cohen's $d = -0.088$), indicating that while the difference is statistically significant (likely due to the large sample size), the **practical significance is limited**. The difference of about 19 minutes may not be clinically meaningful.

Conclusion: While married individuals do report slightly less sedentary time, the difference is minimal from a practical standpoint. This small difference could be attributed to lifestyle factors associated with marriage (e.g., household responsibilities, activities with spouse/family).

Question 3: Effect of Age and Marital Status on Systolic Blood Pressure

Research Question: How do age and marital status affect systolic blood pressure?

Statistical Tests Used

- 1. **Multiple Linear Regression** - To assess the independent effects of age (continuous) and marital status (categorical) on systolic blood pressure (continuous)
- 2. **ANCOVA (Analysis of Covariance)** - To test main effects and examine if there's an interaction

Results

Sample size: 5,838

Descriptive Statistics:

Group	N	Mean SBP (mmHg)	Std Dev
Married	3,186	122.61	18.04
Not Married	2,652	122.87	18.77

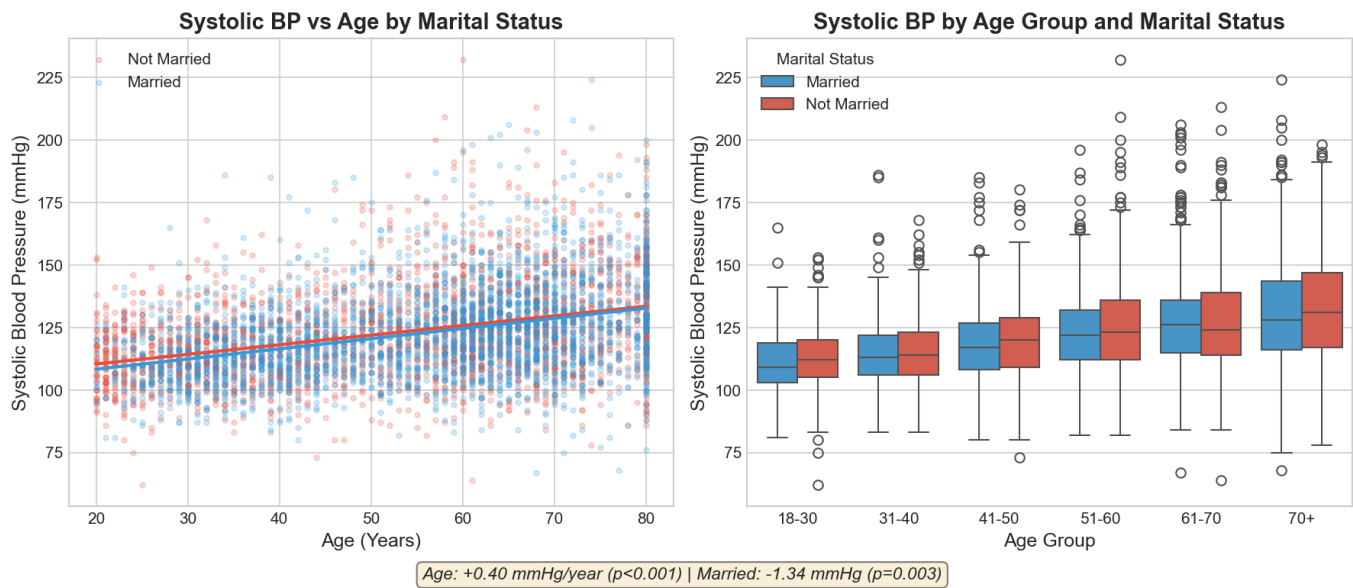
Multiple Linear Regression Output:

Variable	Coefficient	Std Error	t-value	P-value
Constant	102.16	0.764	133.76	< 0.001
Age	0.395	0.013	30.11	< 0.001
Married	-1.342	0.451	-2.98	0.003

- R-squared: 0.134 (13.4% of variance explained)

Interaction Effect: Not significant (p = 0.416), meaning the effect of age on blood pressure is similar for married and non-married individuals.

Visualization



Interpretation

Age Effect ($p < 0.001$):

- For each additional year of age, systolic blood pressure increases by approximately **0.40 mmHg**
- This is consistent with well-established medical literature on age-related increases in blood pressure

Marital Status Effect ($p = 0.003$):

- Married individuals have systolic blood pressure that is about **1.34 mmHg lower** than non-married individuals, after controlling for age
- While statistically significant, this is a relatively small clinical difference

Conclusion: Age is the dominant factor affecting systolic blood pressure, explaining most of the variance. Being married is associated with slightly lower blood pressure, which aligns with research suggesting health benefits of marriage (social support, lifestyle factors). However, the marital status effect, while significant, is modest in magnitude.

Question 4: Correlation between Weight and Sedentary Behavior

Research Question: Is there a correlation between self-reported weight and minutes of sedentary behavior?

Statistical Test Used

Pearson Correlation - This test is appropriate for examining the linear relationship between two continuous variables. Spearman correlation was also calculated for robustness.

Results

Sample size: 6,201

Descriptive Statistics:

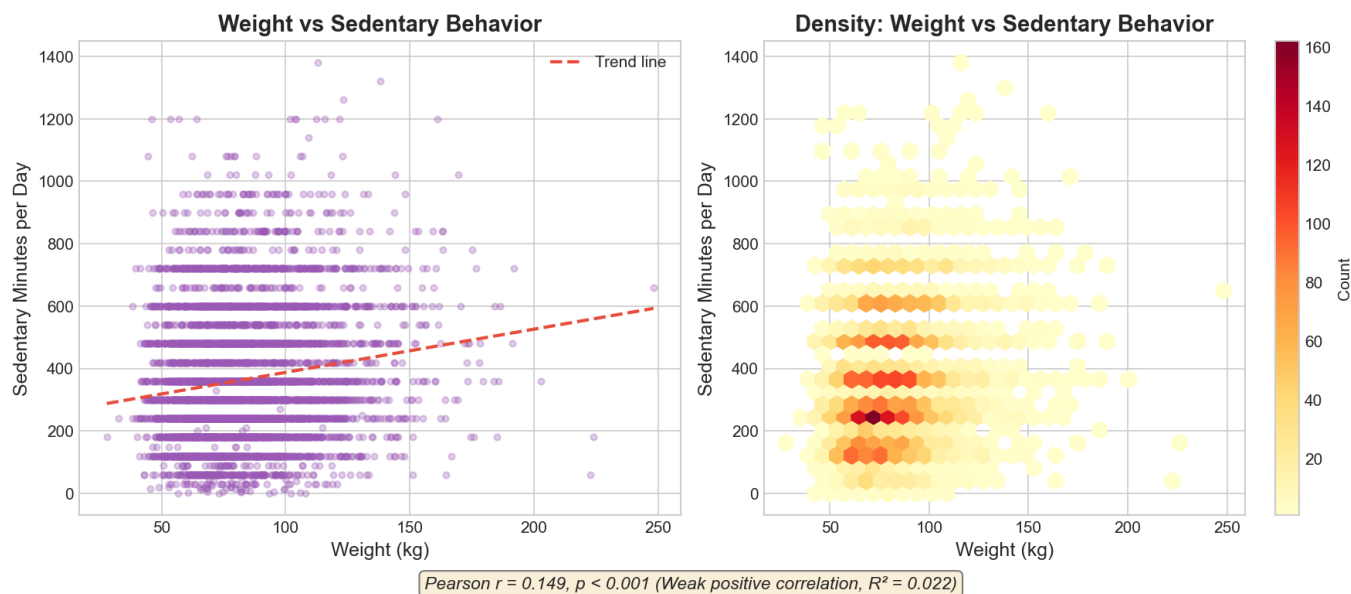
Variable	Mean	Std Dev	Min	Max
Weight (kg)	82.90	22.47	27.9	248.2
Sedentary (min)	364.33	207.93	0	1,380

Correlation Results:

Test	Coefficient	P-value
Pearson r	0.149	< 0.001
Spearman	0.142	< 0.001

- Coefficient of Determination (R^2): 0.022 (2.2% of variance explained)

Visualization



Interpretation

The correlation is **STATISTICALLY SIGNIFICANT** ($p < 0.001$).

There is a **weak positive correlation** ($r = 0.149$) between weight and sedentary behavior. This means:

- Higher weight is associated with more minutes of sedentary behavior
- However, the relationship is weak - weight explains only about 2.2% of the variance in sedentary time

Conclusion: While there is a statistically significant positive relationship between weight and sedentary behavior, the correlation is weak. This suggests that while heavier individuals tend to be slightly more sedentary, many other factors influence sedentary behavior. The bidirectional nature of this relationship should also be considered - sedentary behavior may contribute to weight gain, and higher weight may make physical activity more difficult.

Question 5: Creative Analysis - Systolic Blood Pressure Across Age Groups

Research Question: Is there a difference in systolic blood pressure across age groups (Young Adults 18-39, Middle-Aged 40-59, Older Adults 60+)?

Rationale for Test Selection

I chose **One-Way ANOVA** because:

1. We are comparing means of a continuous variable (systolic blood pressure) across more than two independent groups (three age categories)
2. ANOVA allows us to test whether any group differs from the others
3. Post-hoc tests can then identify which specific groups differ

Results

Sample size: 6,101

Descriptive Statistics by Age Group:

Age Group	N	Mean SBP (mmHg)	Std Dev
Young Adults (18-39)	1,746	112.98	12.62
Middle-Aged (40-59)	1,680	121.37	16.64
Older Adults (60+)	2,675	128.76	19.66

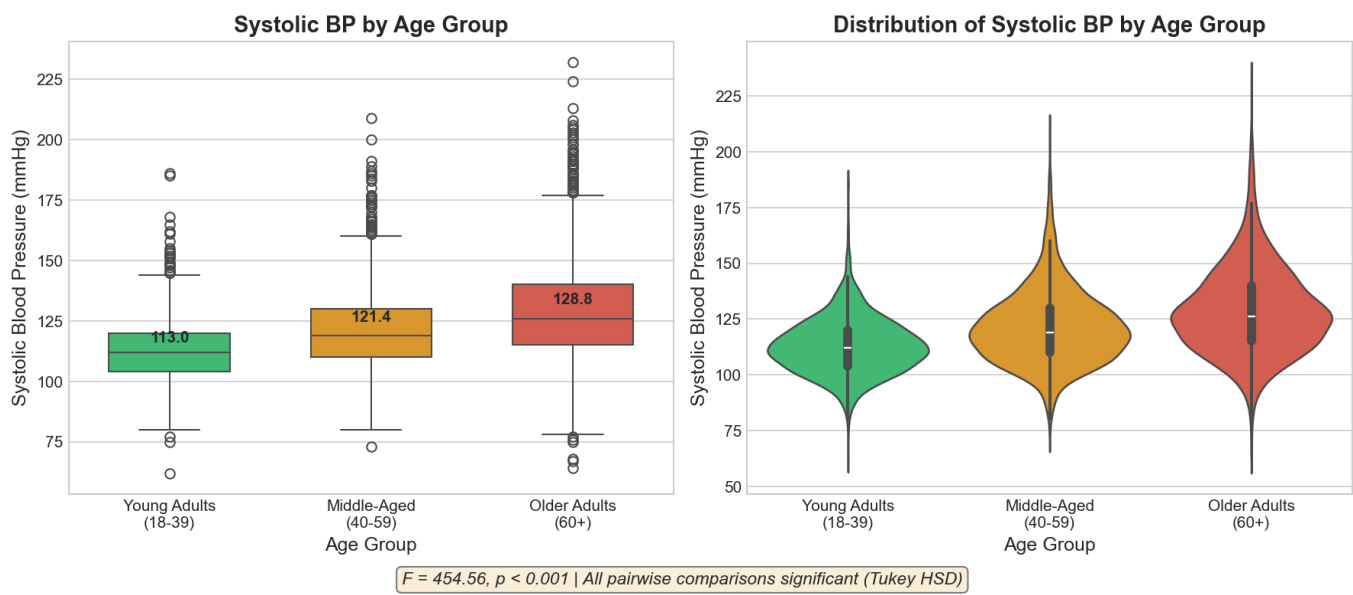
ANOVA Results:

- F-statistic: 454.56
- P-value: < 0.001
- Effect size (Eta-squared): 0.130 (12.97% of variance explained - medium to large effect)

Post-hoc Analysis (Tukey's HSD):

Comparison	Mean Diff (mmHg)	P-value	Significant?
Middle-Aged vs. Older Adults	7.39	< 0.001	Yes
Middle-Aged vs. Young Adults	-8.40	< 0.001	Yes
Older Adults vs. Young Adults	-15.79	< 0.001	Yes

Visualization



Interpretation

We **REJECT** the null hypothesis ($p < 0.001$).

There are **highly significant differences** in systolic blood pressure across the three age groups. All pairwise comparisons are statistically significant:

- Young Adults** have the lowest average systolic BP (112.98 mmHg)
- Middle-Aged** adults have intermediate BP (121.37 mmHg) - about 8.4 mmHg higher than young adults

3. **Older Adults** have the highest average BP (128.76 mmHg) - about 15.8 mmHg higher than young adults

The effect size ($\eta^2 = 0.13$) indicates a **medium to large effect**, meaning age group membership explains a substantial portion of the variation in blood pressure.

Conclusion: Age is a strong predictor of systolic blood pressure, with clear increases across life stages. This finding has important public health implications:

- Young adults generally maintain healthy blood pressure levels
- Blood pressure monitoring becomes increasingly important with age
- Interventions to prevent hypertension should target middle-aged adults before BP elevates further

Summary of All Findings

Question	Test	Key Finding	P-value	Effect Size	Practical Significance
Q1	Chi-Square	Association between marriage & education	< 0.001	-	Moderate
Q2	t-test	Married have less sedentary time	< 0.001	d = -0.09	Small
Q3	Regression	Age increases BP; Marriage slightly lowers BP	< 0.001	R ² = 0.13	Age: Large; Marriage: Small
Q4	Correlation	Weak positive correlation weight-sedentary	< 0.001	r = 0.15	Small
Q5	ANOVA	BP increases significantly across age groups	< 0.001	$\eta^2 = 0.13$	Large

Key Takeaways

1. **Age is the strongest predictor** of systolic blood pressure in this dataset, with clear increases across life stages
2. **Marriage is associated with** higher education levels and slightly better health metrics (less sedentary time, lower BP), though effect sizes are small
3. **Weight and sedentary behavior** are weakly but significantly correlated, suggesting complex relationships between lifestyle factors
4. **Statistical significance vs. practical significance:** Several findings, while statistically significant, have small effect sizes - this is common in large population studies and highlights the importance of interpreting results in clinical context

Analysis conducted using Python with pandas, scipy, and statsmodels libraries.